

Task 2: Evaluating Chaotic Systems

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1 Part A — Phase Space Reconstruction

1.1 Concept

When only a single scalar time series $x(t)$ is available rather than the full state vector, Takens' embedding theorem allows the underlying attractor to be reconstructed by building delay vectors:

$$Y_i = [x_i, x_{i+\tau}, x_{i+2\tau}, \dots, x_{i+(m-1)\tau}] \quad (1)$$

Two parameters must be chosen: the time delay τ and the embedding dimension m .

1.2 Time Delay via Average Mutual Information (AMI)

AMI measures the nonlinear dependence between x_i and $x_{i+\tau}$:

$$I(\tau) = \sum_{i,j} p(x_i, x_{i+\tau}) \cdot \ln \left[\frac{p(x_i, x_{i+\tau})}{p(x_i) \cdot p(x_{i+\tau})} \right] \quad (2)$$

The optimal delay is taken at the first local minimum of $I(\tau)$, which corresponds to the lag at which x_i and $x_{i+\tau}$ share the least redundant information while remaining dynamically related (Fraser & Swinney, 1986).

1.3 Embedding Dimension via False Nearest Neighbours (FNN)

For each candidate dimension d , every point is compared to its nearest neighbour. If that neighbour is only close because the attractor is folded in too few dimensions, the two points will separate sharply when one more coordinate is added. Kennel et al. (1992) define two criteria:

$$\text{Criterion 1: } \frac{|x_{i+d\tau} - x_{j+d\tau}|}{R_d(i)} > R_{\text{tol}} \quad \text{Criterion 2: } \frac{R_{d+1}(i)}{R_A} > A_{\text{tol}} \quad (3)$$

where $R_A = \text{std}(x)$. The optimal m is the smallest dimension at which the FNN percentage drops below a threshold (1%).

1.4 Settings

Table 1: Part A settings.

Setting	Value
System	Lorenz ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$)
Initial condition	(1, 1, 1)
Solver / tolerances	RK45, $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$
dt	0.01
Series length	15 000 samples (after discarding 10 s transient)
AMI config	$\text{max_lag} = 200$, $\text{n_bins} = 32$, $\text{criterion} = \text{first_local_min}$
FNN config	$\text{max_dim} = 10$, $R_{\text{tol}} = 10$, $A_{\text{tol}} = 2$, $\text{threshold} = 1\%$, $\text{theiler} = 50$

1.5 Results

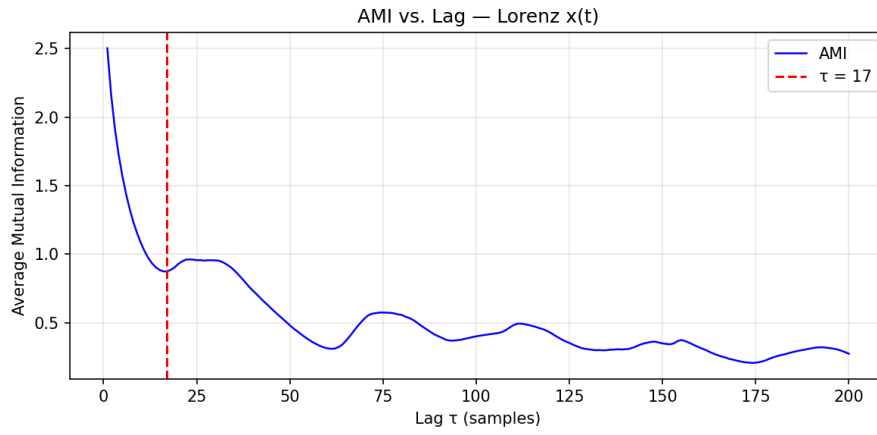


Figure 1: AMI vs. lag for the Lorenz $x(t)$ series. The first local minimum occurs at $\tau = 17$ samples (red dashed line).

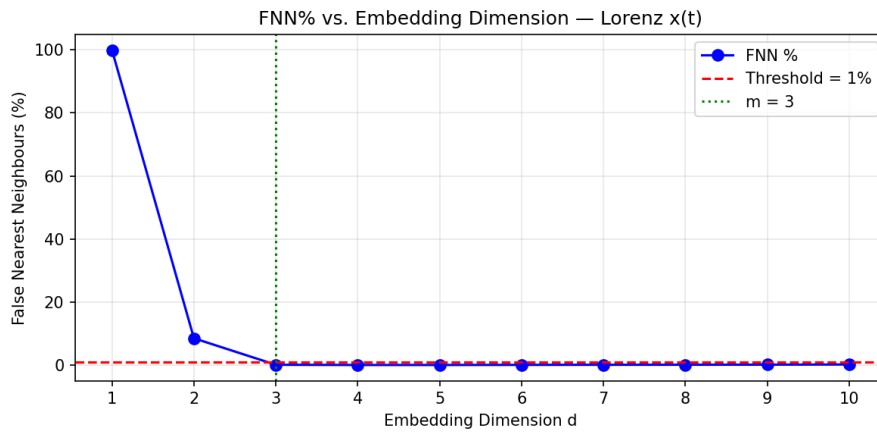


Figure 2: FNN percentage vs. embedding dimension. The percentage drops below the 1% threshold at $m = 3$ (green dotted line).

Table 2: Part A results.

Parameter	Result	Expected range
Time delay τ	17 samples (0.17 s)	10–17
Embedding dimension m	3	3–7

Both values fall within the expected ranges. The embedding dimension $m = 3$ matches the true state dimension of the Lorenz system, and is consistent with Takens’ upper bound $m \leq 2d_A + 1 \approx 5$ (where $d_A \approx 2.05$ is the attractor dimension). The time delay $\tau = 17$ corresponds to 0.17 s, which is roughly one-fifth of the mean orbital period (≈ 0.8 s), placing it well between the over-correlated ($\tau \rightarrow 0$) and decorrelated ($\tau \rightarrow \infty$) extremes.

2 Part B — Lyapunov Exponent

2.1 Concept

The Lyapunov exponent quantifies the average exponential rate of divergence of infinitesimally close trajectories:

$$|\delta Z(t)| \approx |\delta Z_0| \cdot e^{\lambda t} \quad (4)$$

A positive largest exponent $\lambda_1 > 0$ is the defining signature of chaos.

2.2 Method 1 — From the ODE (Wolf’s GSR)

With the governing equations known, the variational equation $\delta \dot{X} = J(X(t)) \delta X$ is integrated alongside the trajectory, where J is the Jacobian:

$$J = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix} \quad (5)$$

Every `ortho_steps` integration steps, the evolved tangent vectors are QR-decomposed; accumulating the log of the diagonal of R and dividing by total time yields the spectrum:

$$\lambda_i = \frac{\sum \ln |R_{ii}|}{T_{\text{total}}} \quad (6)$$

2.3 Method 2 — From a Scalar Time Series (Wolf’s Tracking)

The phase space is reconstructed using τ and m from Part A. Starting from a reference point, its nearest valid neighbour is tracked forward; the growth of their separation is accumulated over many replacement segments:

$$\lambda_1 = \frac{\sum \log_2(d_{\text{final}}/d_0)}{\sum \Delta t_{\text{physical}}} \quad (\text{bits/time}) \quad (7)$$

Conversion to nats/time: $\lambda_1^{(\text{nats})} = \lambda_1^{(\text{bits})} \times \ln 2$.

2.4 Settings

Table 3: Part B settings.

Setting	ODE method	Time-series method
transient_steps	1 000	—
n_steps	30 000	—
ortho_steps	20	—
log_base	e (nats)	2 (bits)
τ, m	—	17, 3 (from Part A)
evolve_cap	—	50
max_replacements	—	500
angle_weight	—	0.5

2.5 Results

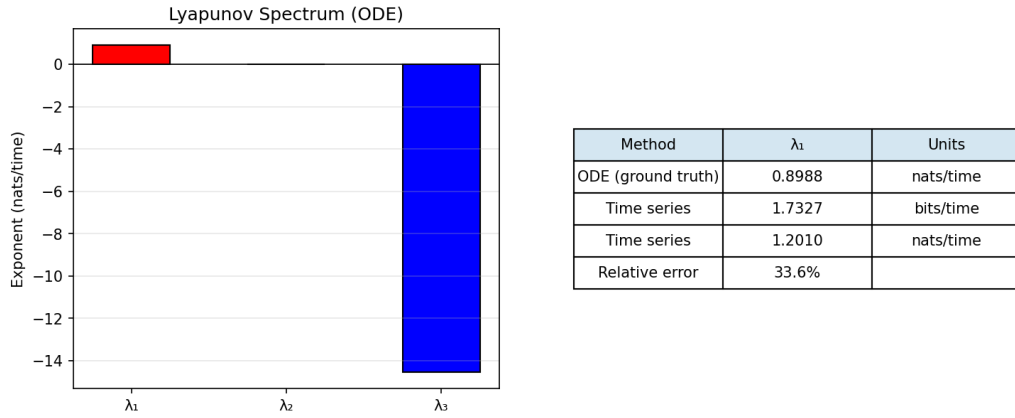


Figure 3: Left: full Lyapunov spectrum from the ODE method. Right: comparison of λ_1 between the two methods.

Table 4: Lyapunov spectrum (ODE method, nats/time).

λ_1	λ_2	λ_3
+0.8988	-0.0023	-14.5547

Table 5: Comparison of λ_1 estimates.

Method	λ_1	Units
ODE (ground truth)	0.8988	nats/time
Time series	1.7327	bits/time
Time series	1.2010	nats/time
Relative error	33.6%	

The ODE spectrum (+0.90, ≈ 0 , -14.55) closely matches the known literature values (+0.91, 0, -14.57), confirming chaotic behaviour: exactly one positive exponent, one near-zero (corre-

sponding to the flow direction), and one strongly negative (contraction onto the attractor). The system is chaotic, not hyperchaotic (which would require two or more positive exponents).

The time-series estimate of $\lambda_1 = 1.20$ nats/time overestimates the ground truth by approximately 34%. This discrepancy is expected for Wolf’s tracking method on finite data: the reconstructed attractor introduces embedding artefacts, the method is sensitive to the choice of neighbour-replacement parameters, and finite series length limits the number of tracking segments available for averaging. Importantly, the time-series method correctly identifies a *positive* λ_1 , confirming chaos from the scalar measurement alone.

3 Part C — Kolmogorov Entropy (K_2)

3.1 Concept

The Kolmogorov–Sinai entropy measures the average rate of information production. Via Pesin’s identity it is bounded by the sum of positive Lyapunov exponents: $K \leq \sum_{\lambda_i > 0} \lambda_i$. A computable lower-bound K_2 is estimated from the correlation sums at consecutive embedding dimensions:

$$K_2 \approx -\frac{1}{\Delta t} \ln \left[\frac{C_m(r)}{C_{m+1}(r)} \right] \quad \text{where } \Delta t = \tau \cdot dt \quad (8)$$

3.2 Results

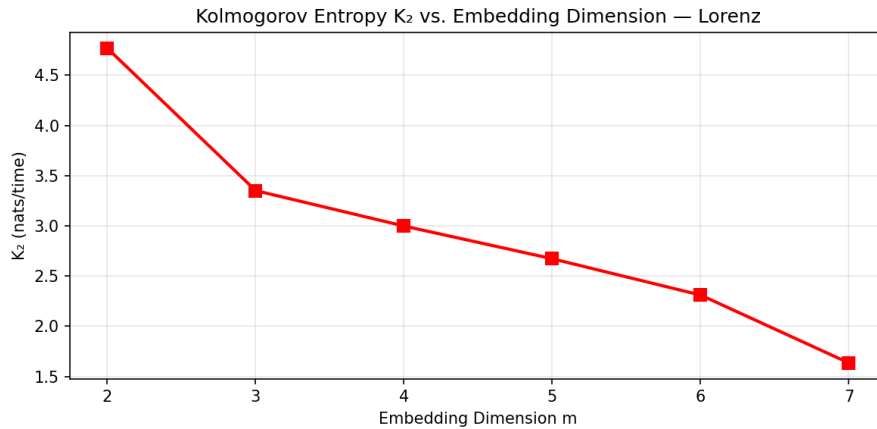


Figure 4: K_2 vs. embedding dimension for the Lorenz system. The entropy is positive and decreases toward saturation as m increases.

Table 6: K_2 estimates at consecutive embedding dimensions ($\Delta t = 0.17$ s).

$m \rightarrow m + 1$	K_2 (nats/time)
$2 \rightarrow 3$	4.77
$3 \rightarrow 4$	3.35
$4 \rightarrow 5$	3.00
$5 \rightarrow 6$	2.67
$6 \rightarrow 7$	2.31
$7 \rightarrow 8$	1.63

K_2 is positive for all embedding dimensions and decreases monotonically, trending toward a finite value as m increases. This convergence to a finite positive value is the hallmark of low-

dimensional deterministic chaos (as opposed to stochastic noise, which would produce K_2 growing without bound). The values are higher than the Pesin bound ($\sum \lambda_i^+ = 0.90$ nats/time), which is expected given the finite-data bias inherent in the correlation-sum estimator and the subsampled pairwise distances.

4 Part D — Correlation Dimension (D_2)

4.1 Concept

The Grassberger–Procaccia algorithm estimates the correlation dimension from pairwise distances:

$$C(r) = \frac{2}{N(N-1)} \sum_{i < j} \Theta(r - \|X_i - X_j\|) \quad (9)$$

The correlation dimension is the scaling exponent:

$$D_2 = \lim_{r \rightarrow 0} \frac{d \ln C(r)}{d \ln r} \quad (10)$$

estimated as the slope of the linear region of a $\log C(r)$ vs. $\log r$ plot.

4.2 Settings

Embedding dimensions $m = 2, 3, \dots, 8$ were tested using $\tau = 17$ from Part A. The radius range was logarithmically spaced (30 values from ~ 0.1 to ~ 30), and a Theiler window of 50 was used to exclude temporally close pairs.

4.3 Results

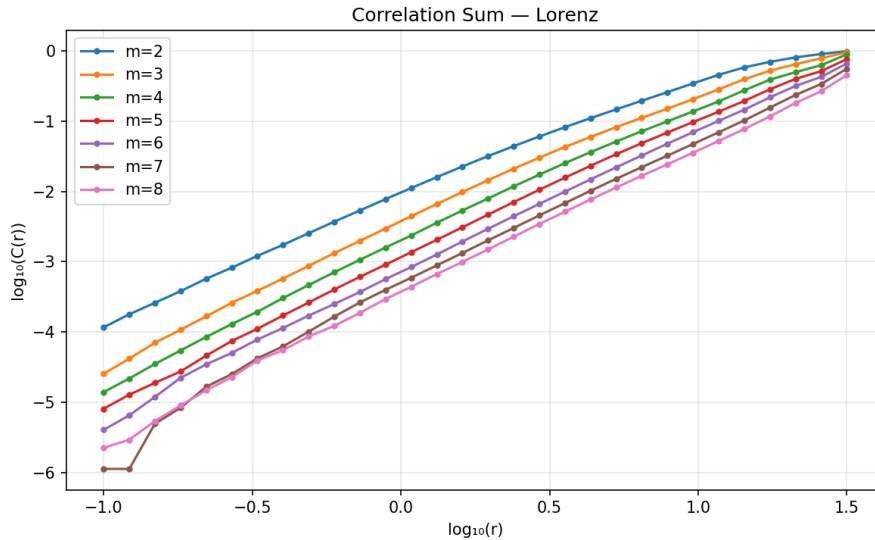


Figure 5: $\log_{10} C(r)$ vs. $\log_{10} r$ for embedding dimensions $m = 2$ through 8. The parallel linear regions confirm consistent scaling behaviour.

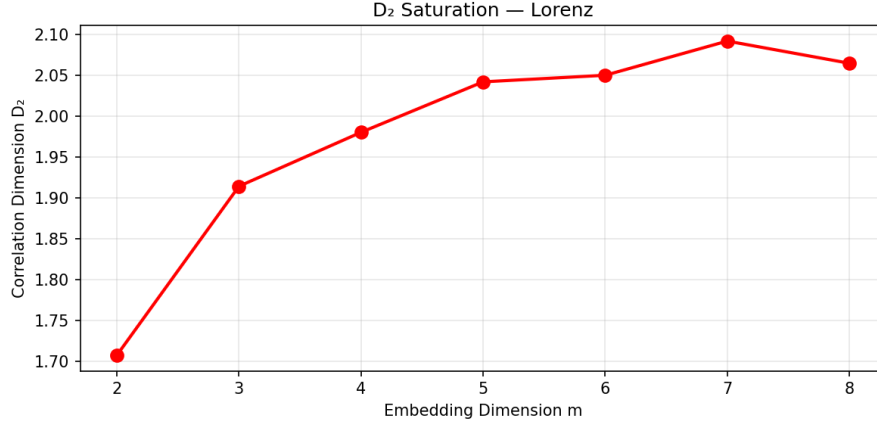


Figure 6: Correlation dimension D_2 vs. embedding dimension m . The value saturates near $D_2 \approx 2.05$ for $m \geq 5$.

The correlation dimension saturates at $D_2 \approx 2.06$, in excellent agreement with the known value of ≈ 2.05 for the Lorenz attractor. The saturation of D_2 as m increases (rather than continuing to grow) confirms that the underlying attractor is finite-dimensional, consistent with low-dimensional deterministic chaos. The non-integer value (≈ 2.05) reflects the fractal geometry of the strange attractor — it is more than a 2D surface but less than a 3D volume.

5 Part E — Box-Counting Dimension (D_0)

5.1 Concept

The box-counting dimension covers the attractor with a grid of boxes of size ε and counts how many boxes $N(\varepsilon)$ contain at least one trajectory point:

$$D_0 = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} \quad (11)$$

5.2 Results

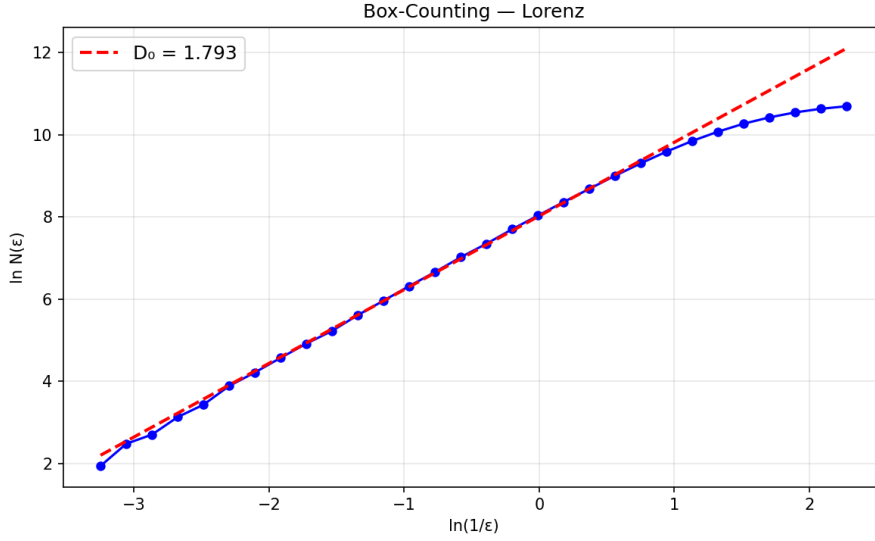


Figure 7: $\ln N(\varepsilon)$ vs. $\ln(1/\varepsilon)$ for the Lorenz attractor. The red dashed line shows the linear fit, giving $D_0 = 1.79$.

Table 7: Dimension estimates comparison.

Dimension	Estimated	Expected
Box-counting D_0	1.79	≈ 2.05
Correlation D_2	2.06	≈ 2.05

The theoretical relationship $D_0 \geq D_2$ holds in the infinite-data limit. With finite data (50 000 points), the box-counting method underestimates the dimension because the smallest boxes are too sparsely populated: the $\ln N(\varepsilon)$ curve saturates at the right end of the plot before reaching the true scaling regime. By contrast, the correlation dimension D_2 is more robust because Grassberger–Procaccia uses pairwise distances (smoother statistics) rather than discrete box counts. Both values are non-integer and consistent with a strange attractor of dimension approximately 2, confirming the fractal nature of the Lorenz system.

6 Conclusion

All five evaluation methods converge on the same conclusion: the Lorenz system at $(\sigma = 10, \rho = 28, \beta = 8/3)$ is **chaotic**.

The Lyapunov spectrum $(+0.90, \approx 0, -14.55)$ shows exactly one positive exponent, classifying the system as chaotic (not hyperchaotic, which would require two or more positive exponents). The time-series-based estimate of λ_1 confirms a positive exponent from the scalar signal alone, validating the phase-space reconstruction.

The Kolmogorov entropy K_2 is finite and positive, converging with increasing embedding dimension — consistent with deterministic chaos and inconsistent with stochastic noise (which would produce unbounded K_2).

The correlation dimension $D_2 \approx 2.06$ is non-integer and saturates, confirming the attractor is a low-dimensional fractal object. The box-counting dimension $D_0 \approx 1.79$ is lower than D_2

due to finite-data limitations but remains non-integer and consistent with a strange attractor of dimension ≈ 2 .

The phase-space reconstruction from a single scalar $x(t)$ using $\tau = 17$ and $m = 3$ faithfully preserves the chaotic properties of the full 3D system, validating Takens' embedding theorem in practice.